

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

27th June 2019

Subject SP2 – Life Insurance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) You work for a recently established life insurance company which specializes in marketing and selling endowment assurances.

- i) Discuss the five key factors which would influence the amount of initial capital needed by the company to write a particular policy contract. (10)

The Chief Actuary of the company has asked you to update a document that highlights the key components that may be used to determine the asset share of a policy. The existing document has listed down premium income, investment income, commission, and expenses as some of items that contribute to the asset share.

- ii) Finish the document by outlining other factors which contribute to the determination of asset share for a policy. (4)
[14]

Q. 2) The regulator of the country Actuarial has recently allowed sale of 2-year individual term products by life insurance companies. Until now, life insurance companies were not allowed to write term life insurance products with term less than 10 years.

A life insurance company operating in the country Actuarial decides to be innovative and launches a single premium 2-year term assurance product with two versions

- a) Base version of the product with base sum assured payable on death.
b) **'Updated version' of the product** with an option to increase the base sum assured, at the end of first year, by 100% by paying normal premium rates at the end of 1st year without any underwriting.

- i) Discuss the risks associated with offering the 'updated version' of the product. (13)
ii) Calculate the additional premium payable by a policyholder at inception who chooses the updated version of product assuming the following:

Base Sum Assured = 10000000

Age at entry = 35 years exact

Normal mortality rates q_x are: $q_{35} = 1.2/1000$, $q_{36} = 1.3/1000$

Kindly note that normal mortality rates are applicable for normal premium rates, for all lives before the option date and for non-takers of the option following the option date.

Mortality rates of policyholder choosing the option after the option date: 150% of normal mortality rates.

Proportion of policyholders who choose the option at the end of first year: 25%.

Interest = 6%

Expenses: Nil (12)
[25]

Q. 3) Term insurance products with options of regular premium payment, limited premium payment and single premium payments are very popular in a country. A life insurer is trying to venture into the term assurance market for the first time.

- i) Describe how it would determine the mortality basis to be used in pricing the term assurance contract. (9)

As per the current product features, there is no surrender value payable in the term insurance products, irrespective of the premium payment term.

The insurance company is considering offering a surrender value in the single premium payment term insurance products.

- ii) Discuss why the company might consider offering surrender values for single premium payment term insurance plans while none are offered for the regular pay option. (3)

- iii) Discuss the methods that can be used to arrive at the surrender values. (6)

- iv) Analyze the suitability of each method for this product. (10)

The insurer is also planning to tie up with service providers of the social networking health application that is a market leader in this field. The service provider will provide health and life style data of their users to the insurance company.

Currently the proposal form is the main source of underwriting information for the company. The insurer is examining the possibility of dropping the health questionnaire from the proposal form and using the information from the service provider instead.

- v) Discuss the risks and other factors to be considered before dropping the health questionnaire from the proposal form. (10)
[38]

Q. 4) An insurance company is selling unit linked insurance contracts for the last 5 years. The products allow the policyholder to choose, and switch between, a variety of funds.

In order to further differentiate its products from other unit linked products sold in the market, the company has designed the products with an investment return guarantee of 3% compounded annually upon maturity.

- i) Suggest possible ways in which the product design could be modified to manage the risk associated with the option. (5)

- ii) Describe the checks it could carry out on the model built to calculate the cost of the guarantee. (8)
[13]

Q. 5) i) State the principles of investment for a life insurance company. (3)

ii) The investment regulations of a country specify the investment norms which are as follows:

- Minimum investment of 50% in State and Government Securities
- Maximum investment of 35% in corporate bonds AA rated and above
- Maximum investment of 15% in equities
- No investment in unquoted equities
- No investment in properties
- No mismatching by currency allowed

Explain the pros and cons of such investment regulations from the perspective of a policyholder.

(7)

[10]
